BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Defensive Design

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does authentication establish?

2. Why translate high-level source code?
   A) The CPU only reads comments
   B) It increases physical RAM
   C) The CPU needs machine code
   D) It removes every logic error

3. "Sequence means the computer chooses any random instruction order." Correct this. Explain why.

4. Insertion sort has sorted part [2, 5, 8] and next value 9. Which values move? Give the new sorted part.

Today's objectives

Explain the core ideas of defensive design using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Anticipate misuse
Reject unexpected input and avoid exposing sensitive detail in errors.

Input sanitisation
Handle or remove harmful/unexpected characters where appropriate.

Maintainability
Meaningful identifiers, comments and modularity make software easier to inspect.

Defensive design
Plans for invalid data, malicious use and foreseeable user mistakes.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Anticipate misuse
Reject unexpected input and avoid exposing sensitive detail in errors.

Input sanitisation
Handle or remove harmful/unexpected characters where appropriate.

Maintainability
Meaningful identifiers, comments and modularity make software easier to inspect.

Defensive design
Plans for invalid data, malicious use and foreseeable user mistakes.

I Do - Worked model

Improve a login counter that accepts unlimited guesses and prints the correct password on failure.

- Remove any display of the secret.
- Limit repeated unsuccessful attempts.
- Present a non-revealing error and lock/slow further attempts.
- Keep attempts in an appropriately scoped variable; test boundary conditions.

We Do - Guided practice

1. A text form permits unescaped SQL input. Suggest improvements.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition that locks an account after three failed attempts.

```
attempts = 3
if ... then
print("Locked")
else
print("Try again")
endif
```

You Do - Independent exam questions

Q2. State two defensive programming techniques. [2 marks]

Q3. Explain why meaningful variable names aid maintainability. [2 marks]

Q4. Explain why showing a user's actual password in an error message is unsafe. [2 marks]

Exit Ticket - True or False?

Input validation is one defensive design technique.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Whether a user has supplied suitable evidence of the claimed identity.

2. C - The CPU needs machine code

3. Instructions execute in the specified order.

4. (a) Values greater than 9 shift right: []. New prefix [2, 5, 8, 9].

I Do - Explain the model

Remove any display of the secret.

Limit repeated unsuccessful attempts.

Present a non-revealing error and lock/slow further attempts.

Keep attempts in an appropriately scoped variable; test boundary conditions.

We Do - Model answers

1. Reject unexpected inputs without leaking data.

Code First - Answer

```
attempts = 3
if attempts >= 3 then
print("Locked")
else
print("Try again")
endif
```

Expected output: Locked

You Do - Self-mark

Q2: Input validation and restricted access or appropriate sanitisation; any two justified techniques.

Q3: They make purpose easier to understand and reduce debugging/maintenance effort.

Q4: It exposes a secret to observers and can enable unauthorised access.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.